

DATE :-

EXPT NO :- 04

INVERTING AND NON INVERTING AMPLIFIER

1. Inverting Amplifier

AIM:-

To design an inverting amplifier using op-amp for the given gain of

-2.

APPARATUS REQUIRED:-

S.NO	components	specification.
1	Operational Amplifier	IC 741
2	Resistors	1K Ω , 2K Ω
3	Breadboard / Trainer kit	1
4	CRO (cathode Ray Oscilloscope)	1
5	Function Generator	1
6	connecting wires	As required.

DESIGN:-

For an inverting amplifier , $A_v = -\frac{R_f}{R_{in}}$

Given ,

$$A_v = -2$$

$$-2 = \frac{R_f}{R_{in}}$$

Taking ,

$$R_{in} = 1K\Omega$$

$$R_f = 2K\Omega$$

output equation

$$V_o = A_v V_{in}$$

$$V_o = -\frac{R_f}{R_{in}} V_{in}$$

Inverting Amplifier circuit

Observation table :-

S. NO	$V_{in}(V)$	$V_{out}(V)$	Practical Gain	Theoretical Gain.
1	1	-2.6	-2.6	-2.2
2	2	-5	-2.5	-2.2
3	3	-7.2	-2.4	-2.2
4	4	-9.4	-2.35	-2.2

PROCEDURE:-

1. Connect IC 741 on the trainer kit
2. Give dual power supply ($+V_{CC}$ and $-V_{CC}$) to the IC
3. Make the circuit connections as per the circuit diagram.
4. Apply sinusoidal input signal from the function generator to the inverting signal.
5. Ground the non-inverting terminal.
6. Connect the feedback resistor (R_f) between output and inverting terminal
7. Switch ON the power supply and CRO
8. Observe the input and output waveforms on the CRO.
9. Verify that the output waveform is amplified and 180° out of phase with the input.
10. Tabulate the readings.

RESULT:-

Thus, the working of the inverting amplifier was studied and verified successfully.

Non-Inverting Amplifier :-

AIM :-

To design a non-inverting amplifier using an op-amp for the given gain of 2.

APPARATUS REQUIRED :-

S. NO	Components	Specifications
1	Operational Amplifier	IC 741
2	Resistors	1K Ω , 2K Ω
3	Function generator	1
4	CRO	1
5	Dual power supply	$\pm 12V$
6	Trainer kit	1
7	Connecting wires	As required

DESIGN :-

For a non-inverting amplifier,

$$V_o = \left(1 + \frac{R_f}{R_{in}} \right) V_{in}$$

gain,

$$A_v = 1 + \frac{R_f}{R_{in}}$$

$$A_v = 2$$

$$2 = 1 + \frac{R_f}{R_{in}}$$

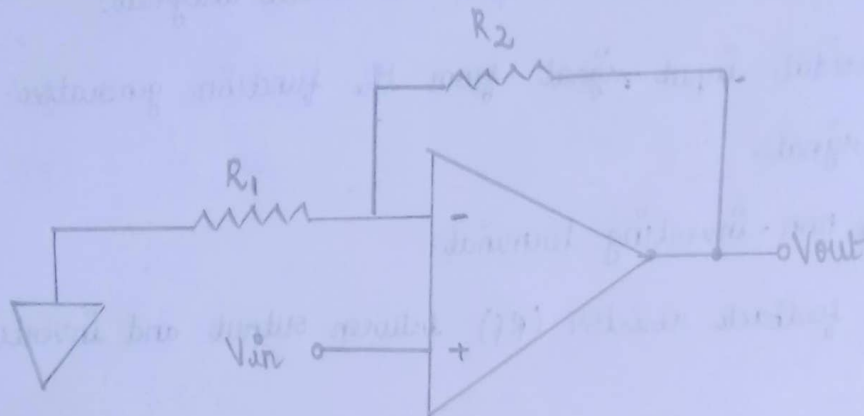
$$R_f = R_{in}$$

Taking,

$$R_i = 1K\Omega$$

$$R_f = 1K\Omega$$

Non-Inverting Amplifier :-



Observation table :-

S.NO	$V_{in}(V)$	$V_{out}(V)$	Practical gain	Theoretical gain
1	0.5V	1.38V	3.76	2.2
2	1V	2.94V	3.94	2.2
3	1.5V	4.73V	4.153	2.2
4	2V	6.26V	4.13	2.2
5	2.5V	7.96V	4.18	2.2
6	3V	9.50V	4.16	2.2
7	3.6V	10.57V	3.936	2.2

PROCEDURE:-

- 1) Place IC 741 on the trainer kit
- 2) Connect dual power supply $\pm 12V$ on to the IC.
- 3) Make connections as per the circuit diagram
- 4) Apply sinusoidal input signal from the function generator to the non-inverting signal terminal
- 5) Connect feedback resistor (R_f) and resistor (R_i) properly.
- 6) Switch on the power supply and CRO
- 7) Observe the input and output voltage waveforms on the CRO.
- 8) Measure the input voltage, output voltage, time period and frequency
- 9) Verify that the output waveform is amplified and inphase with the input waveform
- 10) Tabulate the readings.

RESULT:-

The non-inverting amplifier was designed and verified successfully